



Major Internship: – 24MCA424



PROJECT TITLE

CUSTOMER CHURN & RETENTION ANALYSIS

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Master of Computer Application (MCA)**

By Student Under the Supervision

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CHAPTER 1 — INTRODUCTION

1.1 Introduction

In today's competitive business environment, organizations need to understand their customers in order to maintain long-term relationships and sustainable growth. Customer churn occurs when an existing customer stops using a company's products or services. A high level of customer churn can affect revenue, customer relationships and overall business performance.

Customer data contains valuable information about customer demographics, tenure, services, contracts, payment methods and charges. However, raw customer data cannot directly provide useful business information. It needs to be cleaned, organized, analyzed and visualized.

The Customer Churn & Retention Analysis project uses data analytics techniques to transform raw customer data into meaningful information.

The project combines spreadsheet-based preprocessing, SQL analysis, Python-based exploratory analysis and Power BI visualization into one complete analytical workflow.

1.2 Project Background

Businesses collect customer information through various transactions and services. This information can be used to understand customer behavior.

For example, customer data can help answer questions such as:

- How many customers have left the service?**
- What percentage of customers have churned?**
- How does churn vary across contract types?**
- Does customer tenure have a relationship with churn?**
- How does churn vary according to payment methods?**
- How do monthly charges differ between churned and retained customers?**
- Which customer segments require closer attention?**

The project attempts to answer such questions using structured data analysis.

CHAPTER 2 — PROBLEM STATEMENT

2.1 Problem Statement

Organizations may have large amounts of customer data but may not have an effective analytical process for understanding customer churn.

Manual analysis of customer information can be time-consuming and may make it difficult to identify patterns across multiple customer attributes.

❖ **Therefore, the problem addressed by this project is:**

- **To analyze customer data using SQL, Python and Power BI in order to understand customer churn and retention patterns and generate meaningful business insights.**

The project focuses on understanding existing customer data rather than predicting future churn.

CHAPTER 3 — OBJECTIVES

3.1 Main Objective

The main objective of this project is to analyze customer data and understand patterns associated with customer churn and retention using data analytics techniques.

3.2 Specific Objectives

The project aims to:

- 1. Understand the structure and characteristics of the customer dataset.**
- 2. Clean and preprocess the raw customer data.**
- 3. Store customer data in a MySQL database.**
- 4. Perform SQL-based customer analysis.**
- 5. Analyze customer behavior using Python.**
- 6. Perform Exploratory Data Analysis (EDA).**
- 7. Study relationships between customer attributes and churn.**
- 8. Perform basic statistical analysis.**
- 9. Create meaningful data visualizations.**
- 10. Develop interactive Power BI dashboards.**
- 11. Identify important customer segments and churn patterns.**
- 12. Present analytical results in a business-friendly format.**

CHAPTER 4 — SCOPE OF THE PROJECT

The scope of the project includes:

- **Customer data preprocessing**
- **Data cleaning**
- **Database management**
- **SQL analysis**
- **Exploratory Data Analysis**
- **Customer segmentation**
- **Churn analysis**
- **Retention analysis**
- **Basic statistical analysis**
- **Data visualization**
- **Power BI dashboard development**
- **Business insight generation**

The project is designed primarily for descriptive and diagnostic analysis. Machine learning, deep learning and predictive modeling are not included in the current implementation.

CHAPTER 5 — EXISTING SYSTEM

In a traditional customer analysis system, organizations may rely on spreadsheets, static reports or manual calculations.

Such an approach may create difficulties when dealing with larger datasets.

Limitations of the Existing Approach

- 1. Manual analysis requires considerable time.**
- 2. Static reports provide limited interaction.**
- 3. Identifying relationships between multiple variables can be difficult.**
- 4. Repeated calculations may increase the possibility of errors.**
- 5. Customer segments may not be easily identifiable.**
- 6. Business users may find raw datasets difficult to interpret.**

CHAPTER 6 — PROPOSED SYSTEM

The proposed system provides an integrated data analytics workflow for customer churn and retention analysis.

The workflow is:

Excel → MySQL/SQL → Python → EDA & Statistics → Power BI → GitHub

Each technology is used for a specific purpose.

Excel

Used for initial data inspection and preprocessing.

MySQL

Used for structured storage and SQL analysis.

Python

Used for detailed data analysis.

Pandas and NumPy

Used for data manipulation and numerical operations.

Matplotlib and Seaborn

Used for visualization.

Power BI

Used to develop interactive dashboards.

GitHub

Used to maintain and present the project files and documentation.

CHAPTER 7 — TECHNOLOGY AND TOOLS USED

7.1 Microsoft Excel

Excel is used during the initial stage of the project.

Main Uses

- Dataset inspection
- Sorting and filtering
- Data cleaning
- Identifying missing values
- Initial understanding of customer attributes

7.2 MySQL

MySQL is used as the relational database management system.

Main Uses

- Database creation
- Table creation
- Customer data storage
- Data retrieval
- SQL queries
- Aggregation and filtering
- Business analysis

7.3 SQL

SQL is used to analyze customer data stored in MySQL.

Examples include:

- Customer count
- Churn count
- Retention count
- Churn rate
- Average charges
- Churn by contract
- Churn by payment method
- Churn by services

7.4 Python

Python is used as the primary programming language for analytical work.

It is used for:

- Data loading
- Data cleaning
- Data transformation
- Exploratory analysis
- Statistical analysis
- Visualization

7.5 Pandas

Pandas provides DataFrame-based data manipulation.

It is used for:

- Reading CSV files
- Filtering records
- Handling missing values
- Grouping data

- Aggregation
- Data transformation

7.6 NumPy

NumPy is used for numerical operations and statistical calculations.

7.7 Matplotlib

Matplotlib is used to create visualizations such as:

- Bar charts
- Line charts
- Histograms
- Distribution plots

7.8 Seaborn

Seaborn is used for statistical visualization.

It is used for:

- Count plots
- Heatmaps
- Box plots
- Distribution plots
- Relationship plots

7.9 Jupyter Notebook

Jupyter Notebook provides an interactive environment for executing Python code, displaying visualizations and documenting the analysis.

7.10 Power BI

Power BI is used for business intelligence and dashboard development.

It provides:

- KPI cards
- Charts
- Tables
- Filters
- Slicers

- Interactive visualizations
- Dashboard reporting

7.11 GitHub

GitHub is used to organize and showcase the project.

The repository can contain:

- Dataset
- SQL files
- Python notebook
- Power BI screenshots
- Documentation
- README
- Project resources

CHAPTER 8 — SYSTEM REQUIREMENTS

8.1 Hardware Requirements

- Processor: Intel Core i5 / AMD Ryzen 5 or above
- RAM: Minimum 8 GB
- Recommended RAM: 16 GB
- Storage: Minimum 256 GB SSD
- Standard HD/Full HD display

8.2 Software Requirements

- Windows 10/11
- Microsoft Excel
- MySQL
- Python
- Jupyter Notebook
- Power BI Desktop
- Git/GitHub

8.3 Python Libraries

Pandas, NumPy, Matplotlib, Seaborn

CHAPTER 9 — PROJECT METHODOLOGY

The project follows a structured data analytics lifecycle.

Step 1 — Data Collection

The customer dataset is obtained and loaded into the project environment.

Step 2 — Data Understanding

The dataset is examined to understand:

- Number of records
- Number of columns
- Column names
- Data types
- Customer attributes
- Churn field

Step 3 — Data Preprocessing

The data is checked for:

- Missing values
- Duplicate records
- Incorrect data types
- Invalid values
- Inconsistent values

Step 4 — Excel Analysis

Excel is used for initial inspection, filtering and preprocessing.

Step 5 — MySQL Database

The cleaned customer dataset is imported into MySQL.

Step 6 — SQL Analysis

SQL queries are executed to answer business questions.

Step 7 — Python Analysis

Python is used for:

- Data manipulation
- EDA
- Statistical analysis
- Visualization

Step 8 — Power BI

Power BI is used to create interactive dashboards.

Step 9 — Business Insights

The final analytical results are interpreted from a business perspective.

Step 10 — GitHub Documentation

Project files and documentation are organized for portfolio presentation.

CHAPTER 10 — DATASET DESCRIPTION

The project uses a customer-level dataset containing information about customers and their service relationships.

The Customer Churn & Retention Analysis project uses customer-level data to understand customer behavior, service usage, charges, and churn patterns. The dataset contains important information such as Customer ID, Gender, Senior Citizen, Partner, Dependents, Tenure, Contract Type, Payment Method, Monthly Charges, Total Charges, Internet Service, and Churn Status.

The dataset is used for data cleaning, SQL analysis, exploratory data analysis (EDA), statistical analysis, and Power BI dashboard creation. The main purpose is to identify factors related to customer churn and understand customer retention patterns.

Dataset Type: Customer Service/Telecom Data

Analysis Type: Descriptive and Diagnostic Analysis

CHAPTER 11 — DATA PREPROCESSING

Data preprocessing ensures that the dataset is suitable for analysis.

11.1 Missing Value Analysis

Missing values were checked using Excel and Python.

Python example:

```
df.isnull().sum()
```

This provides the number of missing values in each column.

11.2 Duplicate Analysis

Duplicate records were identified using:

```
df.duplicated().sum()
```

Duplicate records can affect the accuracy of customer-level analysis and therefore need to be examined.

11.3 Data Type Checking

The data types of different columns were examined using:

df.info()

Numeric, categorical and identifier columns were handled according to their requirements.

11.4 Data Validation

The dataset was checked for invalid and inconsistent values before beginning the main analysis.

CHAPTER 12 — MYSQL DATABASE AND SQL ANALYSIS

12.1 Database Creation

A MySQL database was created for storing customer information.

Example:

```
CREATE DATABASE customer_churn_db;
```

12.2 Customer Table

A customer table was created to store customer-level information.

The **Customer ID** acts as the unique identifier for customer records.

12.3 Total Customers

```
SELECT COUNT(*) AS total_customers  
FROM customers;
```

12.4 Churned Customers

```
SELECT COUNT(*) AS churned_customers  
FROM customers  
WHERE Churn = 'Yes';
```

12.5 Retained Customers

```
SELECT COUNT(*) AS retained_customers  
FROM customers  
WHERE Churn = 'No';
```

12.6 Churn Distribution

```
SELECT Churn, COUNT(*) AS customer_count  
FROM customers  
GROUP BY Churn;
```

12.7 Average Monthly Charges

```
SELECT Churn,  
       AVG(MonthlyCharges) AS average_monthly_charges  
FROM customers  
GROUP BY Churn;
```

12.8 Churn by Contract

```
SELECT Contract,  
       Churn,  
       COUNT(*) AS customer_count  
FROM customers  
GROUP BY Contract, Churn;
```

12.9 Churn by Payment Method

```
SELECT PaymentMethod,  
       Churn,  
       COUNT(*) AS customer_count  
FROM customers  
GROUP BY PaymentMethod, Churn;
```

12.10 Average Tenure by Churn

```
SELECT Churn,  
       AVG(Tenure) AS average_tenure  
FROM customers  
GROUP BY Churn;
```

12.11 Overall Churn Rate

```
SELECT  
ROUND(  
    SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END)  
    * 100.0 / COUNT(*), 2  
) AS churn_rate  
FROM customers;
```

CHAPTER 13 — PYTHON DATA ANALYSIS

Python was used to perform detailed analysis of the customer dataset.

13.1 Import Libraries

```
import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

13.2 Load Dataset

```
df = pd.read_csv("customer_churn.csv")
```

13.3 Display Data

```
df.head()
```

13.4 Dataset Shape

```
df.shape
```

This returns the number of rows and columns.

13.5 Dataset Information

```
df.info()
```

13.6 Statistical Summary

```
df.describe()
```

13.7 Missing Values

```
df.isnull().sum()
```

13.8 Duplicate Records

```
df.duplicated().sum()
```

These steps provide an initial understanding of the dataset.

CHAPTER 14 — EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis was performed to identify patterns, distributions and relationships within the dataset.

14.1 Customer Churn Distribution

A count plot was used to compare churned and retained customers.

```
sns.countplot(data=df, x='Churn')  
plt.title('Customer Churn Distribution')  
plt.show()
```

This chart provides an overview of the customer churn distribution.

14.2 Churn by Contract

```
sns.countplot(data=df, x='Contract', hue='Churn')  
plt.title('Customer Churn by Contract Type')  
plt.xticks(rotation=30)  
plt.show()
```

This analysis helps compare churn across contract categories.

14.3 Churn by Payment Method

Payment methods were analyzed to compare customer churn across different payment categories.

14.4 Churn by Internet Service

Internet service categories were compared against customer churn.

14.5 Churn by Tenure

Customer tenure was analyzed to understand differences in customer retention behavior across relationship durations.

14.6 Monthly Charges Analysis

Monthly charges were compared between churned and retained customers.

14.7 Correlation Analysis

A correlation matrix was created for numerical variables.

```
corr = df.select_dtypes(include=np.number).corr()
```

```
sns.heatmap(corr, annot=True)  
plt.title('Correlation Matrix')  
plt.show()
```

Correlation analysis helps identify relationships between numerical variables.

CHAPTER 15 — STATISTICAL ANALYSIS

Basic statistical methods were used to summarize numerical customer information.

15.1 Mean

Mean represents the average value.

```
df['MonthlyCharges'].mean()
```

15.2 Median

```
df['MonthlyCharges'].median()
```

15.3 Standard Deviation

```
df['MonthlyCharges'].std()
```

15.4 Minimum

```
df['MonthlyCharges'].min()
```

15.5 Maximum

```
df['MonthlyCharges'].max()
```

These statistics provide a basic numerical understanding of the customer dataset.

CHAPTER 16 — POWER BI DASHBOARD

Microsoft Power BI was used to create an interactive dashboard based on the analyzed customer data.

The dashboard was designed to present important customer churn and retention metrics in a simple and visually understandable format.

16.1 Dashboard Page 1

Title:

Customer Churn & Retention Overview

KPI Cards

- Total Customers
- Churned Customers
- Retained Customers
- Churn Rate
- Retention Rate

Main Visuals

- Customer Churn Distribution

- Churn by Contract
- Customer Distribution by Contract
- Churn by Payment Method
- Customer Tenure Analysis

Slicers

- Gender
- Contract
- Payment Method
- Internet Service
- Senior Citizen
- Partner
- Dependents

16.2 Dashboard Page 2

Title:

Customer Behavior & Retention Analysis

Main Visuals

- Churn by Tenure
- Churn by Monthly Charges
- Churn by Internet Service
- Churn by Technical Support
- Churn by Online Security
- Churn by Payment Method
- Customer segment summary

The second page provides a deeper analysis of customer characteristics associated with churn and retention.

CHAPTER 17 — RESULTS AND FINDINGS

The completed analysis provides a structured view of customer churn and retention.

The major findings obtained from the analysis include:

1. Customer data can be divided into churned and retained customer groups.
2. Churn distribution varies across different customer categories.
3. Contract type can be compared with customer churn behavior.
4. Customer tenure provides useful information about retention patterns.
5. Payment methods show differences in customer distribution and churn.
6. Monthly charges can be compared between churned and retained customers.
7. Service subscriptions can be analyzed to understand customer behavior.
8. Power BI provides an interactive way to explore these relationships.

The exact numerical values and percentages should be taken directly from the final dataset and Power BI dashboard.

CHAPTER 18 — BUSINESS INSIGHTS

The project provides several business-oriented insights.

18.1 Customer Segmentation

Customers can be grouped according to tenure, contract, payment method and services.

18.2 Churn Monitoring

The churn rate can be monitored through Power BI KPI cards.

18.3 Contract Analysis

Comparing contract categories helps understand differences in customer retention behavior.

18.4 Payment Analysis

Payment methods can be examined to identify differences in churn distribution.

18.5 Service Analysis

Customer service subscriptions such as internet service, online security and technical support can be compared with churn.

18.6 Customer Value Analysis

Monthly and total charges can be analyzed to understand the financial importance of different customer segments.

CHAPTER 19 — ADVANTAGES

The project provides the following advantages:

1. Provides a structured customer analytics workflow.
2. Reduces dependence on manual analysis.
3. Combines SQL and Python analysis.
4. Provides interactive Power BI reporting.
5. Makes customer churn patterns easier to understand.
6. Supports data-driven decision-making.
7. Allows customer segments to be analyzed efficiently.
8. Provides reusable analytical queries and notebooks.
9. Creates a portfolio-ready Data Analyst project.
10. Can be extended with advanced analytics in the future.

CHAPTER 20 — LIMITATIONS

The current project has the following limitations:

1. Analysis depends on the quality of the available dataset.
2. Historical customer data may not represent future behavior.
3. The project focuses on descriptive and diagnostic analytics.
4. Machine learning prediction is not included.
5. Some external factors influencing customer churn may not be available in the dataset.
6. Customer feedback and qualitative information may not be represented.

CHAPTER 21 — FUTURE SCOPE

The project can be extended in several ways.

21.1 Machine Learning

Machine learning algorithms can be added in a future version for churn prediction.

Possible algorithms include:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
-

21.2 Customer Segmentation

Clustering algorithms can be used to create behavior-based customer segments.

21.3 Real-Time Dashboard

The dashboard can be connected to continuously updated data sources.

21.4 Automated Reporting

Automated refresh and reporting can be implemented in Power BI.

21.5 CRM Integration

The analytical system can be connected with CRM systems to support customer retention workflows.

CHAPTER 22 — CONCLUSION

The **Customer Churn & Retention Analysis** project demonstrates the practical application of data analytics techniques for understanding customer behavior.

The project uses a complete analytical workflow beginning with data preprocessing in Excel, structured data analysis using MySQL and SQL, exploratory and statistical analysis using Python, and interactive reporting through Power BI.

The use of Pandas and NumPy makes it possible to manipulate and analyze customer data efficiently, while Matplotlib and Seaborn provide useful visual representations of the data. Power BI brings the analysis together into interactive dashboards that make customer

churn and retention patterns easier to understand.

The project does not use machine learning or predictive modeling. Instead, it focuses on understanding existing customer data through **descriptive and diagnostic analytics**.

Overall, the project demonstrates practical skills required for a Data Analyst role, including data cleaning, SQL, Python, EDA, statistics, visualization, dashboard development and business insight generation.

